

GoApptiv: Improving accessibility to quality medication with cloud- driven technology

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GoApptiv needed
an agile platform
and scale across
beyond, to deliver
medicines to people
in remote locations

About GoApptiv

As one of India's fastest growing end-to-end business service providers in healthcare, GoApptiv is on a mission to improve accessibility to quality medication, particularly for people living in extra urban and rural areas in India. Through its proprietary platform, it plans to build a digital infrastructure that will transform healthcare access in India.

Industries: Transportation & Logistics

Google Cloud results

- Empower vendors in 250,000+ villages across India with genuine medication
- Enable a developer friendly environment with seamless API integration
- Support faster time to market with Cloud Functions
- Build trust with key pharmaceutical companies with Google Cloud security features

In India, access to genuine medication is challenging, particularly for those in remote villages. With low-value transactions, logistic challenges with reach, and the low working capital of smaller chemists in these areas, this has led to

Reduced operational cost with

Location: India

(<https://cloud.google.com/scheduler/>),



About Cloudside Technologies

Cloudside Technologies is driven to help businesses achieve their best business outcomes with cloud as a catalyst. With decades of experience from the team, the Google Cloud partner supports organizations in their cloud native journey, from the discovery phase to launching and scaling.

Technologies

Google Cloud (<https://cloud.google.com/>)

BigQuery (<https://cloud.google.com/bigquery/>)

Cloud Functions

(<https://cloud.google.com/functions/>)

Cloud Run (<https://cloud.google.com/run/>)

Cloud Scheduler

(<https://cloud.google.com/scheduler/>)

Logging (<https://cloud.google.com/logging/>)

Cloud Pub/Sub (<https://cloud.google.com/pubsub/>)

value leakage in the distribution channels, resulting in a less consistent supply of medicines to the rural chemists of India. As the villages get less access to medicine, lookalike products and counterfeit medicines are frequently being sold. Since there aren't many chemists and hospitals in these rural markets, it can be a challenge for people to get access to quality medicines, or verify the authenticity of their prescriptions.

With a mission to make quality medicines available to people in these parts of India, co-founders Sreeram Venkitaraman, Rajasekhar Parcha, and Venkata Ravikanth launched GoApptiv (<https://www.goapptiv.com/>) in 2016 to help connect all stakeholders across the value chain. This is done by using its proprietary technology platforms to deliver the best quality medicines to people, even in the furthest corners of the country.

"India is a big country, and we believe that only through technology, we would be able to solve this pressing issue," explains Ravikanth. Through its app, GoApptiv now supports 250,000 villages by ensuring that chemists and wholesalers have access to quality medicines.

When it was first launched, GoApptiv was on a different cloud provider. "Our initial focus for the first few years was on platform building, so we started on a small scale for a few instances. However, as we continued to build, we realized that we needed more developers," explains Jayendra Gothi, Head of Technology at GoApptiv. As this was still in the early stages, the company needed to keep costs low, so

Gothi and his team began searching for a solution that would allow the DevOps team to quickly build and move forward. "We found Google Cloud (<https://cloud.google.com/>) to be the solution, because of its capability to automate many of the tasks," he says.

Furthermore, as a startup, many of its tech stack were open source, so GoApptiv needed a developer-friendly environment for developers to write APIs outside its systems while remaining globally available. To this end, Google Cloud checked all the boxes.

Building a flexible and agile data warehouse with BigQuery

By 2020, the organization continued to scale, with GoApptiv finding that it had grown past its initial development phase. At this point, Ravikanth and the team needed to focus more on getting insights from its existing data, which was coming through various applications. "We started looking into data warehousing and analysis to improve sales and marketing," explains Gothi. "However, it became difficult to run queries on our SQL servers manually, as the queries would impact the application layers."

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*—Jayendra Gothi, Head of
Technology, GoApptiv*

The team worked with Google Cloud partner Cloudside Technologies (<https://thecloudside.com/>) (Cloudside), who suggested that GoApptiv move its databases to BigQuery (<https://cloud.google.com/bigquery>). In particular, this means using Datastream for BigQuery, which can replicate relational databases directly to BigQuery for real-time insights on operational data. "Initially

we were worried about the amount of time it would take to migrate our data, but we were pleasantly surprised at the speed of Google Cloud, and the fact that we could customize the queries to even more granular areas, making it a real game changer for the company," says Gothi. Since onboarding BigQuery, GoApptiv has reduced its operational costs by up to 20 percent.

Another significant change is the company's ability to get a holistic view of its organizational data. "Prior to the adoption of BigQuery, the dashboards and reports used to connect to multiple databases across different applications. Now, consolidating data from different sources and providing a unified perspective have become a walk in the park. This transformation simplified our data management, while opening avenues for substantial data analytics," says Kishore Battu, Head of Enterprise Architecture at GoApptiv.

On the application level, however, GoApptiv felt that moving fully to Kubernetes from the get go was not the most viable option in terms of costs. With the support of Cloudside, Ravikanth and his team were able to find a workaround on Google Cloud, by leveraging tools like Cloud Run (<https://cloud.google.com/run>), and using virtual machines (VMs) as a container registry with load balancing. "By showing us a workaround, we were able to try manual containerization first so that we can be ready to move completely to an automated state when the time is right for our organization," he says.



Team GoApptiv

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Building a scalable microservices architecture

GoApptiv also relied on Pub/Sub
(<https://cloud.google.com/pubsub>), Cloud Logging
(<https://cloud.google.com/logging>), and Cloud
Scheduler (<https://cloud.google.com/scheduler>) when

deploying its microservices architecture. These solutions have enabled the company to deploy scalable services by eliminating heavy dependency issues. In particular, Pub/Sub has allowed the company to create scalable, event-driven microservices more effectively, relieving them from concerns, such as message queue scalability, message loss, and message retriggering. At the same time, the company implemented the tool for additional cloud functions, such as Google ML Vision, which transforms an image to text, before the message is sent to applications using Pub/Sub.

GoApptiv uses Cloud Logging to centralize log management in real-time, which eliminates the need for log dependencies on the company's application service. "Before implementing Cloud Logging, our applications used to generate logs on a daily basis, which proved to be a cumbersome and challenging process for debugging. We had to rely on scripts for log backups. However, with the introduction of Cloud Logging, managing logs became much more efficient. It allowed us to centralize logs for each application, simplifying the search for specific information, and offered built-in alerting capabilities for specific events," says Sagar Vaghela, Head of Innovation.

Finally, Cloud Scheduler helped GoApptiv reduce its reliance on the application server to run cron tasks, with the tool allowing the company to manage these automation tasks in a single place. Together with the use of Cloud Logging, this also lets GoApptiv better prepare its services for Kubernetes and Cloud Run.

Vaghela shares that the integration of features, such as Pub/Sub, Cloud Scheduler, Cloud Logging, and Cloud Functions, significantly bolstered the resilience of its microservice architecture. "This integration empowered us to establish uniform processes throughout our services, fostering a culture of consistency and efficiency. As a result, innovation becomes more accessible, facilitating rapid implementation."

Speeding up time to market with developer-friendly tools

The team found [Cloud Functions](https://cloud.google.com/functions) (<https://cloud.google.com/functions>) to be a powerful tool for launching new applications, due to its developer-friendly features and automation.

"In a startup environment, short turnaround times are the norm. Clients expect solutions to be ready in less than a month, which is quite a short period of time," explains Ravikanth. "Having tools like Cloud Functions helps make these goals achievable because it removes the need to write an API from scratch and it can be easily integrated where needed." Using Cloud Functions, the DevOps team can develop proof of concepts, using them as interim solutions as the team continues to finetune and deliver the final product.

The adoption of Google Cloud, according to Ravikanth, was a breeze because of its clear documentation. "We could easily experiment with new tools and include products that were a right fit for us at different stages of growth," he says, adding

that working with Cloudside provided some good perspectives and insights for the team.

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—Venkat Ravikanth, Co-
founder and CTO,
GoApptiv

"The Cloudside team certainly helped us in performing one of the critical aspects of our IT goals, and are technically proficient. They managed the project allocated to them within the timelines and allocated budget," says Ravikanth.

Enabling a secure network for all stakeholders

When it comes to ensuring security and regulatory compliance, the pharmaceutical industry has extremely stringent standards. "When we partner with key organizations, the first thing we need to do is fill in a questionnaire, which is primarily made up of data security questions," shares Ravikanth. With Google Cloud, GoApptiv can rest assured that it meets all the compliance guidelines, particularly in the data at rest and data in transit, as these data are already part of the Google hosting requirements. Securing the trust with large pharmaceutical companies is a key battle won, as GoApptiv is able to distribute its medication across the supply chain.

Moving forward, Ravikanth says he has big plans to further explore the CI/CD features of some of the company's critical applications into Kubernetes for

complete containerization. "We're also looking to integrate our existing Google Cloud applications with Google Workspace and the many features it offers, so we can provide a next-level experience to our customers through different communication channels."